

RAPID MVP OPERATING GUIDE

AssetCare AI

Explainable predictive maintenance for government vehicles and assets

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01 - PROJECT AT A GLANCE

One complete decision loop, kept deliberately small

The prototype converts frozen asset, maintenance, and inspection records into a ranked, explained inspection queue. It supports a human decision; it does not replace one.

INPUT	PROCESS	OUTPUT	ACTION
CSV / Excel records	Validate and create time-aware features	0-100 risk score and three reasons	Inspect, monitor, verify, or record disposition

What is complete

- 180-asset deterministic demo dataset with maintenance and inspection history.
- Blocking data checks, model comparison, risk bands, explanations, and action recommendations.
- Overview, priority queue, asset detail, model evidence, data quality, and handoff screens.
- Manual-review safeguard, local disposition log, upload schema validation, and automated tests.

Important data notice

All shipped records and metrics are simulated demo evidence. Do not present them as MINECOFIN or institutional performance results.

02 - RUN LOCALLY

Start from a clean machine in two commands

The project is packaged for Windows PowerShell and runs offline after the initial dependency installation.

Recommended setup

```
powershell -ExecutionPolicy Bypass -File .\setup.ps1
.\run.ps1
```

Open http://localhost:8501 if the browser does not open automatically.

> **What setup does** Creates .venv, installs dependencies, regenerates deterministic demo records, validates them, trains the candidate models, and publishes fresh scores.

Useful maintenance commands

TASK	COMMAND
Regenerate demo data	.\.venv\Scripts\python.exe scripts\generate_demo_data.py
Rebuild scores	.\.venv\Scripts\python.exe scripts\run_pipeline.py
Run tests	.\.venv\Scripts\python.exe -m pytest
Rebuild this guide	.\.venv\Scripts\python.exe scripts\build_guide.py

03 - USER EXPERIENCE

Reach the highest-risk asset and understand it quickly

The primary path is visible immediately. Secondary evidence and data controls stay available without crowding the maintenance decision.

SCREEN	DECISION ANSWERED	KEY CONTENT
Overview	Where should attention go first?	Fleet risk profile, top assets, urgent actions
Priority queue	Which assets match today's review capacity?	Class/band/score filters, reason, last service, action
Asset detail	Why is this asset flagged and what should happen?	Reasons, feature values, timeline, disposition
Model evidence	How was the ranking checked?	Recall, precision, PR-AUC, alert rate, guardrails
Data quality	Can these records be trusted for scoring?	Schema, linkage, plausibility, completeness, upload check

Human-in-the-loop rule

The system recommends inspection priority. Authorized staff confirm the action and record a disposition. Repairs are never authorized automatically.

04 - DATA CONTRACT

Validate identity, labels, and plausibility before modeling

A smaller trustworthy table is better than a large inconsistent one. The pipeline stops on blocking schema, linkage, duplicate-ID, codebook, or plausibility failures.

SOURCE	REQUIRED FIELDS
assets.csv	asset_id, asset_class, make_model, acquisition_date, usage_value
maintenance_events.csv	event_id, asset_id, event_date, event_type, fault_or_defect, severity, maintenance_cost, downtime_hours
inspections.csv	inspection_id, asset_id, inspection_date, defect, severity, outcome

Current frozen-input checks

CHECK	STATUS	DETAIL
Asset schema	Pass	Required fields present
Maintenance schema	Pass	Required fields present
Inspection schema	Pass	Required fields present
Unique asset IDs	Pass	0 duplicate asset IDs
Maintenance linkage	Pass	100.0% linked to a valid asset ID
Inspection linkage	Pass	100.0% linked to a valid asset ID
Event codebook	Pass	Event types valid
Plausibility	Pass	0 invalid usage values; 0 invalid costs
Critical completeness	Pass	98.0% complete across critical asset fields
Corrective label signal	Pass	34.8% of events are corrective or breakdown

05 - FEATURES AND MODEL

Simple, comparable, and explainable

Monthly snapshots use only information known at each scoring date. A future corrective or breakdown event within 60 days is the supervised target.

FEATURE FAMILY	EXAMPLES
Recency	Days since service, corrective repair, and inspection
Frequency	Repairs in 90/180 days and repeated fault count
Severity	Maximum recent inspection or corrective-event severity
Usage	Class-aware usage since last service and missing flag
Cost / impact	Recent corrective cost and downtime hours
Context	Asset age, class, and stale-inspection flag

Current simulated holdout evidence

SELECTED	RECALL	PRECISION	PR-AUC
Random forest	**80.0%**	**19.9%**	**0.322**

MODEL	RECALL	PRECISION	PR-AUC	ALERT RATE
Weighted rules	45.7%	22.5%	0.227	31.7%
Logistic regression	80.0%	18.5%	0.268	67.1%
Random forest	80.0%	19.9%	0.322	62.7%

06 - RISK, REASONS, AND ACTIONS

A score must lead to an understandable next step

The displayed 0-100 value is an operational ranking score, not a calibrated probability of failure. Fixed bands keep the demo deterministic.

BAND	SCORE	DEFAULT ACTION
Critical	80-100	Inspect within 24 hours; restrict use if a safety defect is confirmed
High	60-79.9	Inspect within 48 hours and verify the reported signal
Watch	35-59.9	Review within seven days and monitor the next cycle
Normal	0-34.9	Continue scheduled maintenance and monitor monthly
Manual review	No score	Verify identity and missing records before scoring

Current top five demo assets

ASSET	CLASS	SCORE	TOP REASON
GOV-0099	Sedan	88.6	1 corrective repair in 90 days
GOV-0067	Pickup	87.4	2 corrective repairs in 90 days
GOV-0164	Sedan	87.3	critical defect severity recorded recently
GOV-0111	Pickup	86.8	critical defect severity recorded recently
GOV-0011	Pickup	86.4	high defect severity recorded recently

07 - QUALITY AND RELEASE

Break the demo before the judges do

Release checks focus on broken joins, future leakage, impossible scores, contradictory reasons, empty results, and application crashes.

AREA	MINIMUM EVIDENCE
Data	No duplicate IDs; valid linkage, dates, usage, costs, and event codes
Model	Fixed feature order, time-based split, bounded output, baseline comparison
App	Clean launch, working filters, sorted queue, matching asset detail
Explanations	Reason text agrees with underlying values
Fallbacks	Sparse assets show manual review; empty filters show a helpful state
Reproducibility	One setup command, one run command, frozen input, regenerated scores

Automated evidence

The current suite covers blocking data gates, future-event exclusion, sparse history, band boundaries, explanation consistency, score bounds, queue ordering, and manual review.

08 - FIVE-MINUTE LIVE DEMO

Move from public-sector pain to a visible action

Keep the interaction deterministic and use one high-risk example with readable evidence.

1. Problem	Fixed schedules underuse historical maintenance and inspection records.
2. Data	Show the simulated-data badge, required fields, and passed data gates.
3. Overview	Point out the risk distribution and the highest-priority assets.
4. Asset	Open a Critical asset and explain the score with two or three signals.
5. Action	Record an inspection request and reinforce human authorization.
6. Evidence	Compare rules, Logistic Regression, and Random Forest.
7. Close	Describe the approved-data validation sprint and controlled pilot.

09 - TEAM AND CHANGE DISCIPLINE

Work in parallel without integration chaos

The score-output schema is the shared contract. Material changes must update the code, tests, README, status, changelog, and this PDF together.

ROLE	ACCOUNTABILITY
Lead / PM	Scope, issues, integration, decision log, presentation story
Data / ML	Data contract, cleaning, labels, features, models, metrics
App / UX	Screens, filters, queue, asset detail, disposition flow
Domain / QA	Maintenance logic, explanations, action wording, release evidence

Project structure

```
app.py
src/minicofin_pm/{data,pipeline,scoring,ui}.py
scripts/{generate_demo_data,run_pipeline,build_guide}.py
data/{raw/demo,processed}/
models/risk_model.joblib
tests/
output/pdf/
```

10 - BOUNDARIES AND NEXT STEP

A credible prototype ends with an honest handoff

Operational impact targets require a controlled pilot. The rapid MVP proves technical feasibility and decision usefulness only.

IN THIS MVP	DEFERRED
File ingestion and validation	Live IFMIS integration
Batch risk scoring and reasons	Real-time telemetry or hardware retrofits
Human inspection recommendation	Automated repair authorization
Local demo audit trail	Enterprise identity, hosting, and monitoring
Simulated model evidence	Downtime and cost-reduction claims

Next evidence

Approved source fields, real data volume, label quality, maintenance-SME review, operational baseline, and user feedback.

Next engineering

Approved integration path, authentication, security, monitoring, support, and deployment only after the validation sprint.

Definition of ready

The team can move from a frozen input file to a visible, explained maintenance-risk decision in a repeatable local demo.